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SOEN 6481 Systems Requirements Specification

NourishU

Healthy Living Made Simple and Affordable

Requirements Evaluation and Risk Analysis

Delivery #2

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No name or student ID is included in this report, consistent with anonymous submission requirements.

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1. Introduction and Evaluation Scope

The NourishU Vision Document, version 1.1, is assessed in this report. Inconsistencies, conflicts, alternative conflict resolutions, quantitative selection of resolutions, and risks that could keep the project from meeting its goals for healthy living, cost, safety, privacy, usability, and delivery are the main topics of the evaluation.

The Delivery #2 task sequence is followed in the analysis. Task 1 documents the results of the inspection. In a Kotonya and Sommerville interaction matrix, Task 2 records the most important interacting statements. Conflict-resolution operators are used in Task 3. In Task 4, options are chosen using weighted matrices. Task 5 uses a checklist, component inspection, and a risk tree to identify risks; quantitatively evaluates likelihood and severity; and uses risk reduction leverage (RRL) to assess risk countermeasures.

1.1 Evaluation Conventions

- Conflict matrix value 0: statements are distinct and do not overlap.
- Conflict matrix value 1000: statements overlap but do not conflict.
- Conflict matrix value 1: statements conflict under all conditions or under an identified boundary condition.
- Weighted option score = sum of (criterion weight x option score). Criterion weights sum to 1.00.
- Risk exposure (RE) = likelihood x severity. Severity uses the five-level odd-number scale: 1 = Low, 3 = Medium, 5 = High, 7 = Very High, and 9 = Extreme High.
- RRL = (RE before countermeasure - RE after countermeasure) / relative countermeasure cost.

All probabilities, severities, option scores, and effort costs in this draft are reasoned estimates for requirements analysis. They should be validated with the sponsor, a nutrition professional, developers, and representative students before baselining.

2. Task 1 - Inspection and Identification of Inconsistencies

The Vision Document underwent a thorough inspection, examining each section meticulously. The inspection addressed terminology, designation, and structural conflicts; strong or weak discrepancies; ambiguity; incompleteness; non-verifiability; and the absence of traceability between stakeholder responsibilities and product features.

2.1 Inspection Record

ID	Location	Defect type	Severity	Inspection finding	Recommended correction
F01	Sections 4.1 and 4.2	Designation clash	Major	NourishU is referred to as a self-contained application; however, it relies on essential functions such as grocery pricing, nutrition, and email services. The definition of self-contained is ambiguous.	Define self-contained as ownership of the core application and database, while explicitly identifying external-service-dependent functions.
F02	Sections 3.1, 3.2, and 5.12	Role/designation clash	Major	Self-contained refers to having ownership of the core application and database. The Nutrition Advisor is identified as a non-user stakeholder, whereas the Content Moderator is a system user linked to the Nutrition Advisor. The distinctions between review, approval, and publishing authority are not clearly defined, while clearly identifying functions that depend on external services.	Define distinct roles: The Nutrition Advisor is responsible for validating nutrition rules; the Content Moderator oversees the content workflow; only individuals in authorized roles are permitted to publish.
F03	Sections 3.3, 5.1, and 5.2	Structure clash	Major	The user environment describes shared households consisting of two to five individuals, whereas the product model is designed around a single student profile, a specific calorie target, and a defined set of restrictions.	Limit release 1 to a single student profile and permit only serving-size scaling; postpone multi-person nutrition planning.
F04	Sections 5.3 and 4.2	Weak conflict	Critical	The plan is reported to align with the weekly budget; however, the cost is merely an estimate derived from external, cached, or default data. Actual checkout prices may surpass the budget.	Utilize the most recent prices, show the timestamp and confidence range, and implement a configurable contingency buffer.
F05	Section 6 - Availability	Weak conflict	Major	The service must be accessible at all times, although scheduled maintenance is permitted. Without a boundary, it is not always possible for both statements to be satisfied.	Replace the definitive statement with a quantifiable availability target, along with a clear maintenance exclusion and notification period.
F06	Sections 4.1 and 4.2	Weak conflict	Major	The core experience is reported to stay operational during external outages, whereas the budget, calorie, and grocery features rely on external data. Core experience remains undefined.	Define degraded-mode functions, cached-data age limits, and specify which operations are unavailable during an outage.
F07	Sections 5.6 and 7.2	Ambiguity/weak conflict	Critical	The system excludes declared allergens from all plans and recipes but also flags allergens present in related items. It is unclear whether incompatible items can still be recommended or displayed.	Hard-exclude incompatible items from personalized results; warnings may appear only in non-personalized browsing or substitution explanations.
F08	Sections 5.5, 5.6, and 5.12	Incomplete safety requirement	Critical	Halal and allergen compliance does not address cross-contamination, ambiguous ingredients, substitutions, certification expiry, or unknown metadata.	Add provenance, review status, expiry, substitution revalidation, and unknown-data handling requirements.
F09	Section 5.1	Scope/safety ambiguity	Critical	The profile can contain diabetes and low-sodium conditions, which introduces health-sensitive personalization without defining whether the product provides medical advice or how such guidance is validated.	State that NourishU is not a medical device; restrict outputs to general nutrition guidance and require qualified review of health-related rules.
F10	Section 6	Non-verifiable wording	Major	Terms such as quickly, a few steps, around the clock, short as possible, current browsers, and good practices have no fit criteria.	Add measurable response-time, task-step, availability, maintenance, browser-version, and accessibility criteria.
F11	Sections 3.2, 5.13, and 6	Omission/traceability gap	Major	The System Administrator is tasked with monitoring, updates, backups, performance, and security; however, the feature list primarily focuses on accounts, roles, and permissions.	Add capabilities for administration, audit, backup/restore, monitoring, and incident management, or revise the outlined responsibilities.
F12	Sections 4.2, 5.3, and 5.9	Incomplete data requirement	Major	Price freshness, geographic coverage, package size, taxes, discounts, store differences, and unit conversion are not specified.	Define price-source coverage, timestamp, normalization, fallback, and uncertainty behavior.
F13	Sections 5.1 and 5.2	Incomplete behavior	Major	The document does not define behavior when required profile fields are missing or constraints are mutually incompatible and no feasible meal plan exists.	Validate profile input, identify incompatible constraints, explain why no plan is feasible, and allow the user to revise non-safety preferences.
F14	Sections 3.2, 5.13, and 6	Access-control ambiguity	Critical	Personal data is accessible to the owner or authorized roles, but authorized roles and permitted fields are not enumerated. Support and content staff access is unclear.	Define a role-permission matrix, least-privilege access, privileged-user MFA, audit logs, and field-level restrictions.
F15	Sections 5.8 and 5.12	Workflow omission	Major	Recipe publication, review status, versioning, re-review after edits, and withdrawal of unsafe content are not defined.	Introduce draft/reviewed/published/retired states, review ownership, change history, and automatic expiry or re-review.

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ID	Location	Defect type	Severity	Inspection finding	Recommended correction
F16	Section 6 - Security and Privacy	Bundled requirement	Minor	Authentication, encryption, authorization, and privacy are integrated into a single high-level requirement, which complicates traceability and subsequent verification.	Separate the statement into distinct security and privacy requirements, each with unique identifiers and corresponding fit criteria.

Inspection result: 16 findings were recorded: 5 critical, 10 major, and 1 minor finding. The highest-priority issues concern allergen and health safety, data privacy, strict budget claims based on uncertain prices, and missing behavior for incompatible inputs.

3. Task 2 - Interaction Matrix for Documented Conflicts

Ten statements were selected because they participate in the most consequential conflicts. The wording below preserves the intent of the Vision Document while giving each statement an identifier for interaction analysis.

ID	Statement
S1	Each weekly meal plan aligns with the student's food budget for the week.
S2	Meal costs are estimated using external supermarket data, with default or cached data serving as a backup.
S3	All customized plans and recipes do not include declared allergens.
S4	Related products that include allergens are explicitly marked.
S5	The service is accessible 24/7.
S6	Scheduled maintenance may disrupt the service when communicated beforehand.
S7	The core experience continues to operate effectively even when an external service is not accessible.
S8	Budget, calorie, and grocery functions rely on external pricing and nutrition data.
S9	The system assists students who sometimes organize meals for households consisting of two to five individuals.
S10	Each meal plan and recipe suggestion is tailored based on an individual student's profile, budget, dietary restrictions, and calorie goals.

3.1 Kotonya-Sommerville Interaction Matrix

Statement	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	Total
S1	0	1	0	0	0	0	0	1000	0	1000	2001
S2	1	0	0	0	0	0	0	1000	0	0	1001
S3	0	0	0	1	0	0	0	0	0	1000	1001
S4	0	0	1	0	0	0	0	0	0	1000	1001
S5	0	0	0	0	0	1	1000	0	0	0	1001
S6	0	0	0	0	1	0	1000	0	0	0	1001
S7	0	0	0	0	1000	1000	0	1	0	0	2001
S8	1000	1000	0	0	0	0	1	0	0	0	2001
S9	0	0	0	0	0	0	0	0	0	1	1
S10	1000	0	1000	1000	0	0	0	0	1	0	3001
Total	2001	1001	1001	1001	1001	1001	2001	2001	1	3001	14010

The overall total amounts to 14,010. Integer division results in 14 non-conflicting overlap entries, along with a remainder of 10 conflict entries. Due to the symmetry of the matrix, there are 7 distinct non-conflicting overlaps and 5 unique conflict pairs.

3.2 Conflict Interpretation and Boundary Conditions

Conflict	Statements	Type	Boundary condition	Reason
C1	S1 vs. S2	Weak conflict	Actual prices, taxes, package sizes, or outdated fallback values may vary from the estimate.	A strict budget guarantee and an uncertain estimate cannot both be satisfied.
C2	S3 vs. S4	Designation/weak conflict	A related item presented as a recommendation includes a stated allergen.	Exclude and flag indicate different intended result sets; however, those sets remain undefined.
C3	S5 vs. S6	Weak conflict	A student accesses NourishU during scheduled maintenance.	Constant availability throughout the day contradicts the allowance for downtime.
C4	S7 vs. S8	Weak conflict	A pricing or nutrition service is not available when a new plan or grocery list is requested.	The core functionality remains undefined, and numerous visible features rely on external services.
C5	S9 vs. S10	Structure/weak conflict	Household members possess varying calorie targets, allergies, religious guidelines, or financial constraints.	A single personal profile cannot adequately represent a multi-person household.

4. Task 3 - Conflict Resolution

Alternative resolutions were produced by softening statements, circumventing boundary conditions, focusing on the source or target of the conflict, reestablishing compatibility following a boundary event, or eliminating a lower-priority scope statement. New requirements have been introduced that necessitate more than just a simple wording change.

4.1 Candidate Resolutions and Revised Requirements

Candidate-requirement prefixes indicate their subject area: BR = Budget, SA = Safety/Allergen, AV = Availability, EX = External-service continuity, and HH = Household support. The number distinguishes alternative requirements within each subject area.

Option	Conflict	Operator/tactic	Resolution concept	Candidate requirement
C1-A	C1	Weaken conflicting statement	Replace the budget guarantee with an estimate-based condition.	BR-1: The system shall present an estimated weekly cost derived from the most recent normalized prices and shall clearly indicate that this estimate does not serve as a guarantee of checkout pricing.
C1-B	C1	Avoid boundary condition	Block plan generation whenever current live prices are unavailable.	BR-2: The system shall produce a budget-constrained plan only when all necessary price records are no older than 24 hours.
C1-C	C1	Avoid boundary + weaken	Use estimate, timestamp, uncertainty range, and a contingency buffer.	BR-3: The system shall maintain the estimated basket cost at or below 90% of the specified budget by default, display the price timestamp and source coverage, and issue a warning when confidence is low.
C2-A	C2	Specialize conflict target	Separate personalized recommendations from general catalog browsing.	SA-1: Personalized plans and recommendations must not include any items that contain a declared allergen. General search may show incompatible recipes solely when they are distinctly labelled as unsuitable and cannot be included in the plan.
C2-B	C2	Avoid risk/boundary	Hide all incompatible recipes and substitutions from the student.	SA-2: The system shall completely exclude any recipe or substitute that has a positive or unknown allergen status for a declared allergen.
C2-C	C2	Weaken exclusion	Allow incompatible items with warnings.	SA-3: The system might present incompatible recipes even if a prominent allergen warning is displayed prior to selection.
C3-A	C3	Weaken absolute statement	Define a measurable service level and maintenance exception.	AV-1: NourishU will attain a minimum of 99.5% monthly availability, excluding scheduled maintenance that does not exceed two hours per month; notification will be given at least 48 hours in advance.
C3-B	C3	Avoid boundary condition	Adopt zero-downtime deployment and redundant infrastructure.	AV-2: Planned releases must ensure uninterrupted student access by utilizing rolling deployment and redundant instances.
C3-C	C3	Drop lower-priority precision	Retain around-the-clock wording with a general disclaimer.	AV-3: NourishU should typically be accessible at all times, except during necessary maintenance periods.
C4-A	C4	Specialize target + restore	Define cached degraded mode and recover data refresh later.	In the event of an external data outage, students will be able to access their profiles, saved plans, and cached recipes. New cost and nutrition calculations will display data age and refresh automatically following service recovery.
C4-B	C4	Avoid boundary condition	Maintain a fully mirrored internal pricing and nutrition dataset.	EX-2: NourishU shall maintain an internal continuously synchronized copy of all external records required for plan generation.
C4-C	C4	Weaken core-function claim	Disable dependent features while preserving authentication and saved data.	EX-3: In the absence of required external data, the system will disable new calculations for plans, costs, nutrition, and groceries, and will provide an explanation of this temporary limitation.
C5-A	C5	Specialize conflict source	Create separate household profiles with per-person constraints.	HH-1: A household plan must include a profile for each member, addressing their allergens and dietary restrictions while also calculating portions and shared costs.
C5-B	C5	Weaken household scope	Support serving-size scaling, not household nutrition personalization.	HH-2: Release 1 will produce nutrition guidance tailored for a single student profile and will enable the scaling of recipe and grocery quantities from one to five servings.
C5-C	C5	Drop lower-priority statement	Remove household support from the first release.	HH-3: Release 1 shall support meal planning for one student only.

5. Task 4 - Conflict Evaluation with Weighted Matrices

Each set of alternatives is assessed independently, as the alternatives for various conflicts cannot be directly compared. Scores vary from 0.00 to 1.00, with 1.00 indicating that the option completely meets the criterion. The chosen option has the highest weighted score, prioritizing safety constraints over cost when required.

C1 - Budget accuracy versus uncertain prices

Criterion	Weight	C1-A Estimate + disclaimer	C1-B Live data; block if stale	C1-C Estimate + buffer + timestamp
Conflict elimination	0.35	0.75	0.95	0.90
Budget reliability	0.30	0.70	0.90	0.90
Feasibility/cost	0.20	0.95	0.45	0.80
Transparency/usability	0.15	0.85	0.60	0.90
Weighted total	1.00	0.790	0.782	0.880

Selected option: C1-C. It gives the best balance of affordability, feasibility, and transparency without preventing use whenever a live price is unavailable.

C2 - Allergen exclusion versus allergen flagging

Criterion	Weight	C2-A Separate incompatible-results section	C2-B Hard-exclude incompatible items	C2-C Warning only
Safety	0.45	0.90	1.00	0.55
Conflict elimination	0.30	0.90	1.00	0.45
Usability	0.15	0.75	0.65	0.95
Implementation effort	0.10	0.80	0.90	1.00
Weighted total	1.00	0.868	0.938	0.625

Selected option: C2-B. Safety receives the greatest weight. Unknown or positive allergen status is treated as incompatible, eliminating unsafe personalized output.

C3 - Around-the-clock availability versus maintenance

Criterion	Weight	C3-A Measurable SLA + maintenance window	C3-B Zero-downtime deployment	C3-C Vague 24/7 wording + disclaimer
Availability clarity	0.35	0.90	1.00	0.50
Feasibility/cost	0.30	0.95	0.45	1.00
User continuity	0.20	0.75	1.00	0.60
Maintainability	0.15	0.90	0.70	0.80
Weighted total	1.00	0.885	0.790	0.715

Selected option: C3-A. It is testable and realistic for a student-focused pilot while allowing controlled maintenance.

C4 - External-service outage versus core continuity

Criterion	Weight	C4-A Cached de-graded mode	C4-B Full internal mirror	C4-C Disable dependent features
Continuity	0.35	0.85	0.95	0.45
Data quality	0.30	0.80	0.95	0.90
Feasibility/cost	0.20	0.85	0.40	0.95
Transparency	0.15	0.95	0.80	0.90
Weighted total	1.00	0.850	0.817	0.753

Selected option: C4-A. It preserves useful functions, exposes stale data clearly, and is less costly than a complete mirror.

C5 - Household use versus single-student profile

Criterion	Weight	C5-A Full household profiles	C5-B Serving-size scaling only	C5-C Remove household support
Scope feasibility	0.35	0.45	0.90	1.00
Student need	0.30	0.95	0.80	0.45
Nutrition safety	0.20	0.85	0.90	0.95
Simplicity	0.15	0.45	0.90	1.00
Weighted total	1.00	0.680	0.870	0.825

Selected option: C5-B. It addresses common shared cooking and shopping without pretending that one calorie target safely represents several people.

5.1 Consolidated Conflict Decisions

Conflict	Selected option	Decision
C1	C1-C	Estimated cost with contingency buffer, source coverage, timestamp, and low-confidence warning.
C2	C2-B	Hard-exclude positive or unknown allergen items from personalized results and substitutions.
C3	C3-A	Use a measurable 99.5% monthly availability target with bounded announced maintenance.
C4	C4-A	Provide cached degraded mode with explicit data age and automatic refresh after recovery.
C5	C5-B	Limit release 1 to one student profile while scaling servings and grocery quantities.

6. Task 5 - Risk Management

Risk management adheres to the identify-assess-control cycle. Product and process risks were identified through the application of all three techniques mandated by the assignment: a risk checklist, component inspection, and a risk tree. The resulting risk register is subsequently evaluated quantitatively and managed through risk-reduction strategies and RRL.

6.1 Risk Identification by Checklist

Checklist category	Checklist question	Risks found
Functional completeness	Can valid user inputs produce no feasible plan or an incomplete grocery list?	R6
Data quality	Can nutrition, allergen, halal, ingredient, or price data be inaccurate, stale, or incomplete?	R1, R2, R3
Safety	Could the system recommend an allergen, contraindicated ingredient, or misleading health guidance?	R2
Security/privacy	Could an unauthorized party read or modify dietary, religious, allergy, health, or budget data?	R4
Availability/reliability	Can cloud, database, or external-service failures make critical functions unavailable or corrupt saved data?	R5
Usability/accessibility	Could users misunderstand profile questions, warnings, costs, or nutrition summaries?	R7
Content governance	Can unreviewed, expired, or incorrectly edited content be published?	R9
Process/scope	Can the number of features and integrations exceed the available course-project time and resources?	R8

6.2 Risk Identification by Component Inspection

Component	Failure mode	Possible causes	Consequences	Risk IDs
User Profile	Missing, conflicting, or outdated allergy, budget, or health-condition data	Optional fields, unclear validation, user error	Unsafe or infeasible plan; poor personalization	R2, R6, R7
Meal Plan Generator	No feasible plan or plan violates a hard constraint	Constraint interaction or algorithm defect	Student cannot use the service or receives unsuitable meals	R2, R6
Recipe Recommendation	Restricted ingredient is suggested	Filter defect, substitution, unknown metadata	Allergic reaction or religious/dietary violation	R2
Calorie and Budget Engine	Incorrect nutrition or cost totals	Bad units, stale data, rounding, taxes/discounts omitted	Misleading nutrition guidance or budget overrun	R1, R3
Grocery List Generator	Quantities are duplicated, omitted, or not scaled	Recipe aggregation or serving conversion defect	Waste, shortage, or extra spending	R3, R6
Content Management	Unreviewed or expired content is published	Workflow bypass or moderation backlog	Unsafe/inaccurate recipes remain visible	R2, R9
Authentication and RBAC	Unauthorized access or privilege misuse	Weak authentication, broad roles, missing audit	Disclosure or alteration of sensitive personal data	R4
Application Database	Loss, corruption, or unavailable records	Operational error, outage, failed backup	Loss of profiles/plans and service interruption	R5
External Nutrition Data	Unavailable or inaccurate values	Provider outage, schema change, stale records	Incorrect calorie and nutrient calculations	R1, R5
External Grocery Price Data	Stale, incomplete, or geographically irrelevant prices	Provider coverage and refresh limits	Budget estimate differs materially from actual cost	R3, R5
Web User Interface	Warnings or profile questions are misunderstood	Poor wording, accessibility defect, mobile layout issue	Incorrect inputs, abandonment, unsafe choices	R7
Development Process	Delivery scope exceeds available time	Fourteen features, integrations, and safety controls	Late or incomplete delivery and reduced quality	R8

6.3 Risk Tree

The chosen root failure for the risk tree is an unsafe or unsuitable meal recommendation. The OR node indicates that any significant cause can lead to the root failure. The decomposed metadata branch reveals an AND condition where both defective labelling and failed governance play a role in contributing to unsafe published content.

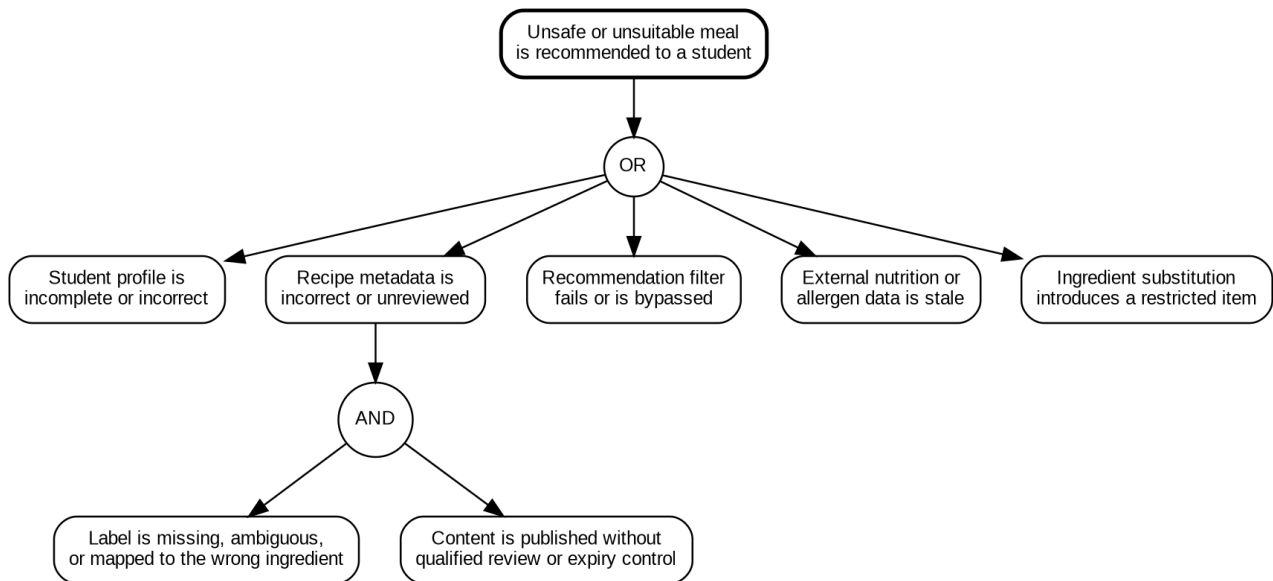


Figure 1: Risk tree for unsafe or unsuitable meal recommendation.

6.4 Consolidated Risk Register and Quantitative Assessment

ID	Type	Risk event	Likelihood	Likelihood rationale	Severity (1, 3, 5, 7, 9)	RE	Priority
R1	Product - data quality	Nutrition or calorie values are incorrect or use inconsistent units.	0.35	Several external and manually curated sources may use different serving sizes and update cycles.	7	2.45	High
R2	Product - safety	A personalized plan contains an allergen, religiously prohibited item, unsafe substitution, or misleading health-related guidance.	0.20	Hard filters and review are intended, but metadata gaps, substitutions, and user-input errors remain credible.	9	1.80	High*
R3	Product - affordability	The actual grocery basket exceeds the student's budget.	0.60	Prices vary frequently by store, location, tax, package size, promotions, and source freshness.	5	3.00	High
R4	Product - security/privacy	Sensitive profile, religious, health, allergy, or budget data is exposed or changed without authorization.	0.20	The system is web-based, stores sensitive fields, and defines several privileged roles.	9	1.80	High*
R5	Product - availability/reliability	Cloud, database, or external-service failure interrupts planning or causes data loss.	0.30	Multiple external services and shared cloud storage create several failure points.	7	2.10	High
R6	Product - functional	No feasible plan is generated, or grocery quantities are incomplete because constraints conflict.	0.35	Budget, calories, allergies, cuisine, time, skill, and religious rules can make the solution space empty.	5	1.75	Medium
R7	Product - usability/accessibility	A student misunderstands profile fields, warnings, nutrition information, or cost uncertainty.	0.45	Users have varied technical and cooking skills and may use small mobile screens.	5	2.25	High
R8	Process - delivery	The project misses schedule or quality objectives because the scope and integrations are too large.	0.55	The Vision Document includes fourteen features plus external services, safety, privacy, moderation, and administration.	7	3.85	Very High
R9	Product/process - content governance	Review backlog or weak workflow leaves inaccurate or expired content published.	0.40	Qualified review is a dependency, while the number and update frequency of recipes may grow quickly.	7	2.80	High

Severity scale: 1 is Low; 3 is Medium; 5 is High; 7 is Very High; and 9 is Extreme High. **Priority rule:** $RE \leq 1.00$ is Low; $1.00 < RE \leq 2.00$ is Medium; $2.00 < RE \leq 3.00$ is High; and $RE > 3.00$ is Very High. A severity of 9 raises the risk to at least High even when exposure is 2.00 or lower. The asterisk marks this severity override.

6.5 Risk Control Alternatives and RRL

Risk	Alt.	Countermeasure	Tactic	Residual P	Residual S	Residual RE	Cost points	RRL	Decision
R1	A	Validate units/ranges, store source metadata, and perform periodic nutrition-advisor sampling.	Reduce risk likelihood	0.12	5	0.60	4	0.463	Select
R1	B	Use one authoritative nutrition database only.	Avoid/reduce risk	0.10	5	0.50	6	0.325	Not selected
R2	A	Hard allergen/religious filter, unknown-data block, substitution re-check, and automated safety tests.	Avoid risk consequence	0.06	9	0.54	5	0.252	Select; combine with re-view
R2	B	Human pre-publication review without an automated hard block.	Reduce risk likelihood	0.10	9	0.90	4	0.225	Secondary control
R3	A	Use timestamped prices, normalized units, a 10% contingency buffer, and low-confidence warning.	Reduce consequence likelihood	0.30	5	1.50	2	0.750	Select
R3	B	Purchase real-time multi-store pricing with broad local coverage.	Reduce risk likelihood	0.15	5	0.75	7	0.321	Defer
R4	A	Encrypt data, enforce least-privilege RBAC, privileged-user MFA, and audit logs.	Reduce risk likelihood	0.07	9	0.63	6	0.195	Select
R4	B	Add data minimization, field-level encryption, and isolated sensitive-data storage.	Avoid/mitigate consequence	0.04	7	0.28	8	0.190	Future hardening
R5	A	Cache read-only data, provide degraded mode, automate backups, and test restoration.	Mitigate consequence	0.12	5	0.60	4	0.375	Select
R5	B	Deploy multi-region active redundancy for the complete stack.	Avoid consequence	0.05	5	0.25	9	0.206	Defer
R6	A	Validate constraint combinations, explain infeasibility, and let users relax only non-safety preferences.	Reduce risk likelihood	0.12	3	0.36	3	0.463	Select
R6	B	Escalate every infeasible case to a support specialist.	Restore/mitigate	0.20	5	1.00	4	0.188	Backup only
R7	A	Use an accessible profile wizard, plain-language warnings, usability testing, and input confirmation.	Reduce risk likelihood	0.20	3	0.60	4	0.413	Select
R7	B	Provide a help center without redesigning the input workflow.	Mitigate consequence	0.35	5	1.75	2	0.250	Not selected
R8	A	Define an MVP and defer lower-priority integrations/features to later releases.	Avoid risk boundary	0.25	5	1.25	2	1.300	Select
R8	B	Add developers and a project manager while retaining full scope.	Reduce risk likelihood	0.30	5	1.50	8	0.294	Not selected
R9	A	Use draft-review-publish-retire states, review SLA, expiry, audit history, and automatic withdrawal.	Reduce risk likelihood	0.15	5	0.75	4	0.513	Select
R9	B	Add moderators without changing the workflow.	Reduce risk likelihood	0.20	5	1.00	6	0.300	Not selected

The interpretation of the RRL indicates that alternatives are assessed within the same risk framework, as the cost points represent relative effort estimates. A higher RRL results in greater exposure reduction for each unit of effort expended. For R2, safety warrants the implementation of layered controls, despite the automated hard block being the primary choice. For R4, the higher-reduction architecture continues to be a viable option for future hardening, even though it has a slightly lower RRL.

6.6 Selected Risk-Countermeasure Requirements

Requirement ID	Selected countermeasure requirement
CM-R1	The system shall normalize nutrition units and serving sizes, store the source and update date for every nutrition record, reject values outside configured validation ranges, and flag records due for qualified review.
CM-R2	The system shall prevent a recipe or substitute from entering a personalized plan when any declared allergen is present or when required allergen or religious-compliance metadata is unknown.
CM-R3	The system shall calculate budget estimates using timestamped normalized prices, apply a configurable contingency buffer of 10% by default, and display source coverage and uncertainty.
CM-R4	The system shall encrypt sensitive data in transit and at rest, enforce least-privilege role-based access, require multi-factor authentication for privileged roles, and record access to sensitive fields.
CM-R5	The system shall provide degraded read-only access to profiles, saved plans, and cached recipes during external-data outages and shall back up application data daily with a quarterly restore test.
CM-R6	The system shall validate profile constraints before plan generation and, when no feasible plan exists, identify the conflicting constraints without automatically relaxing allergies or religious restrictions.
CM-R7	At least 90% of representative student participants shall complete profile setup and weekly-plan generation without assistance during usability testing; all critical warnings shall meet applicable accessibility contrast and keyboard-use criteria.
CM-R8	Release 1 shall prioritize profile management, safe recipe recommendation, single-student weekly planning, budget estimation, and grocery-list generation; advanced tracking, ratings, email notifications, and full household planning may be deferred.
CM-R9	Recipe content shall follow draft, review, published, and retired states; publication shall require an authorized reviewer; edits to safety-relevant fields shall trigger re-review; and expired content shall be removed automatically from personalized results.

6.7 Residual Risk Summary

Risk	Initial RE	Residual RE	Reduction	Residual treatment
R1	2.45	0.60	1.85	Accept with monitoring
R2	1.80	0.54	1.26	Accept with monitoring
R3	3.00	1.50	1.50	Additional review required
R4	1.80	0.63	1.17	Accept with monitoring
R5	2.10	0.60	1.50	Accept with monitoring
R6	1.75	0.36	1.39	Accept with monitoring
R7	2.25	0.60	1.65	Accept with monitoring
R8	3.85	1.25	2.60	Additional review required
R9	2.80	0.75	2.05	Accept with monitoring

The selected controls reduce every residual exposure to 1.25 or lower. R8 remains above 1.00 because schedule risk cannot be eliminated by requirements alone; the MVP scope decision must be enforced through change control. R2 also remains safety-critical despite low residual exposure and therefore requires continuous monitoring, test coverage, content review, and incident response.

7. Conclusions and Recommended Vision Document Changes

The Vision Document establishes a coherent student-centered concept, but several high-level statements are too absolute for the stated dependencies and safety context. The most important revisions are to distinguish estimated cost from guaranteed budget compliance, make allergen handling fail-safe, define measurable availability and degraded mode, limit release 1 to single-student nutrition with serving-size scaling, clarify privileged roles and content-review authority, and introduce explicit behavior for infeasible profiles.

The risk assessment identifies scope overrun and price uncertainty as the highest-exposure risks, while allergen safety and privacy require priority treatment because of their severity. The selected countermeasures should be incorporated into the next Vision revision and traced into use cases, supplementary requirements, design decisions, and test cases in Delivery #3.

References

- [1] NourishU Vision Document, version 1.1, Delivery #1, July 2026.
- [2] SOEN 6481 Delivery #2 assignment instructions, Requirements Evaluation and Risk Analysis, Summer 2026.
- [3] SOEN 6481 Lecture 3, Part 1: Requirements Evaluation and Risk Analysis - inconsistency management, interaction matrices, conflict-resolution operators, risk identification, assessment, control, and RRL.
- [4] SOEN 6481 Lecture 3, Part 2: Requirements Evaluation and Risk Analysis - weighted matrices and quantitative decision support.
- [5] G. Kotonya and I. Sommerville, interaction-matrix technique, 1997, as cited in the course materials.
- [6] A. van Lamsweerde, Requirements Engineering: From System Goals to UML Models to Software Specifications, Wiley, 2009.

A. Task 0 Activity Log

Complete this table with the actual dates and time personally spent. Do not submit estimated or fabricated hours. The total must match the work recorded in the version history or document-editing platform where possible.

Date	Task/activity	Time spent
18-07-2026	Task 0 - Set up activity log and report structure	1 hr.
18-07-2026	Task 1 - Inspect Vision Document and complete inspection records	3 hr.
18-07-2026	Task 2 - Build and interpret interaction matrix	2 hr.

Continued on next page

Date	Task/activity	Time spent
19-07-2026	Task 3 - Generate conflict resolutions and revised requirements	2.5 hr.
19-07-2026	Task 4 - Define criteria, score alternatives, and select resolutions	2 hr.
24-07-2026	Task 5 - Identify, assess, and control risks; calculate RRL	3 hr.
24-07-2026	Final review, anonymization, formatting, and submission preparation	5 hr.
	Total	18.5 hrs.

B. Traceability from Findings to Decisions

Inspection findings	Conflict	Selected resolution	Revised requirement	Risk/countermeasure trace
F04	C1	C1-C	BR-3	R3 / CM-R3
F07-F09	C2	C2-B	SA-2	R1, R2, R9 / CM-R1, CM-R2, CM-R9
F05	C3	C3-A	AV-1	R5 / CM-R5
F01, F06	C4	C4-A	EX-1	R1, R3, R5 / CM-R1, CM-R3, CM-R5
F03	C5	C5-B	HH-2	R6, R8 / CM-R6, CM-R8
F02, F11, F14-F16	Supporting revisions	N/A	Role, security, administration, and workflow clarifications	R4, R5, R9 / CM-R4, CM-R5, CM-R9